Causal Inference -- Job Training Program

Generated by uber-polya

2026-02-26

Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Causal Inference

Universe

- Naive: \{estimate, bias\}
- Propensity Score Matching: \{att, propensity_treated_mean, propensity_control_mean, error\}
- Difference In Differences: \{estimate, error\}
- Doubly Robust: \{ate, error\}

Variables

Symbol	Meaning	Type / Range
x	decision variable	\mathbb{{R}}

Structure

- **Treated**: 500 employees who participated in the training program (approximately)
- **Control**: 500 employees who did not participate (approximately)
- **Covariates**: education_years, experience_years, age (all affect both participation and salary)
- **Confounding**: Higher-education, more-experienced employees self-select into training AND earn more
- **Question**: What is the causal effect of the training program on salary?
- **Ground truth**: The data is synthetic with a true treatment effect of $5,000

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

(1) Selection bias**: Treated and control groups differ systematically on covariates, inflating n

(2) Propensity score**: P(treatment | covariates) *summarizes confounding into a single numb*

(3) ATE vs ATT**: Average Treatment Effect (population) vs Average Treatment Effect on the T

(4) Confounding**: When a variable affects both treatment assignment and outcome

(5) Doubly robust estimation**: Combines propensity model with outcome model; consistent if

Approach

Suggested approach: Naive + Propensity Score Matching + Difference-in-Differences + Doubly

Complexity class:

Available tools: Naive + Propensity Score Matching + Difference-in-Differences + Doubly

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	Naive + Propensity Score Matching + Difference-in-Differences + Doubly
Time:	0.0094s

Solution Details

Solution computed successfully.

Verification

✓	Naive Biased	Passed
✓	Psm Close	Passed
✓	Did Close	Passed
✓	Dr Close	Passed
✓	Psm Closer Than Naive	Passed
✓	Dr Closer Than Naive	Passed
✓	Propensity Overlap	Passed
✓	Treatment Balance	Passed
✓	All Passed	Passed

3. Interpretation

The Question

- **Treated**: 500 employees who participated in the training program (approximately)
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- **Ground truth**: The data is synthetic with a true treatment effect of $5,000.

The Answer

A feasible solution was found.

What This Means

- Naive difference: Overestimates the effect (well above $5,000) due to selection bias
- Propensity Score Matching (ATT): Recovers approximately $5,000
- Difference-in-Differences: Recovers approximately $5,000
- Doubly Robust (AIPW): Recovers approximately $5,000
- Verification: All 8 checks pass -- naive is biased, causal methods are closer to truth

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio